

Roll No

EX-8001 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Computer-Aided Design of Electrical Machine
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Develop an algorithm for optimization of linear problem.
b) Enumerate general approach for optimization.
2. Explain various methods of computer aided design of electrical machine. Explain non-linear programming method and discuss the advantages of computer aided design of electrical machines.
3. a) Explain the design procedure for design of field winding of DC machine.
b) Explain the procedure for calculation of main dimension of DC machine.
4. Explain the detail procedure for armature design of DC machine.
5. Develop the algorithm for detail design of transformer.
6. Why do the turbo alternators have cylindrical rotor instead of salient pole rotor. Determine for a 12.5 MVA, 6.6kV, 2 pole, 50Hz, 3 ϕ alternator suitable values for diameter at air gap, length of core, number of stator slots and conductors. Assume approximate values for missing design constants.
7. a) What are factors are to be considered in selecting the number of rotor slots in case of stator connected slip ring induction motor.
b) Explain the phenomenon of logging and crawling in induction motor. What steps will you suggest at the design to minimize the occurrence of this phenomenon.
8. Explain the following (Any two):
 - i) Design of end rings
 - ii) Linear and non-linear programming
 - iii) Optimal design of induction motor
